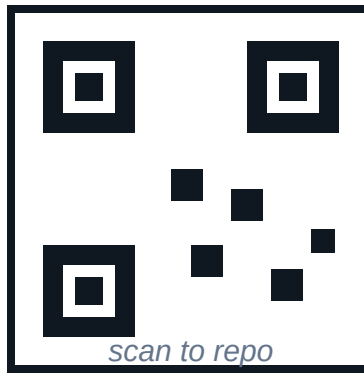


Statistical Screening for Poisoned Machine-Learning Training Data



Detection and Quarantine of Suspicious Samples at the Data-Ingestion Boundary

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Security Engineering • AI / Agent Security • Cloud Security

MIT • Python 3.12 • HEAD b45d662 • verified 2026-09-26

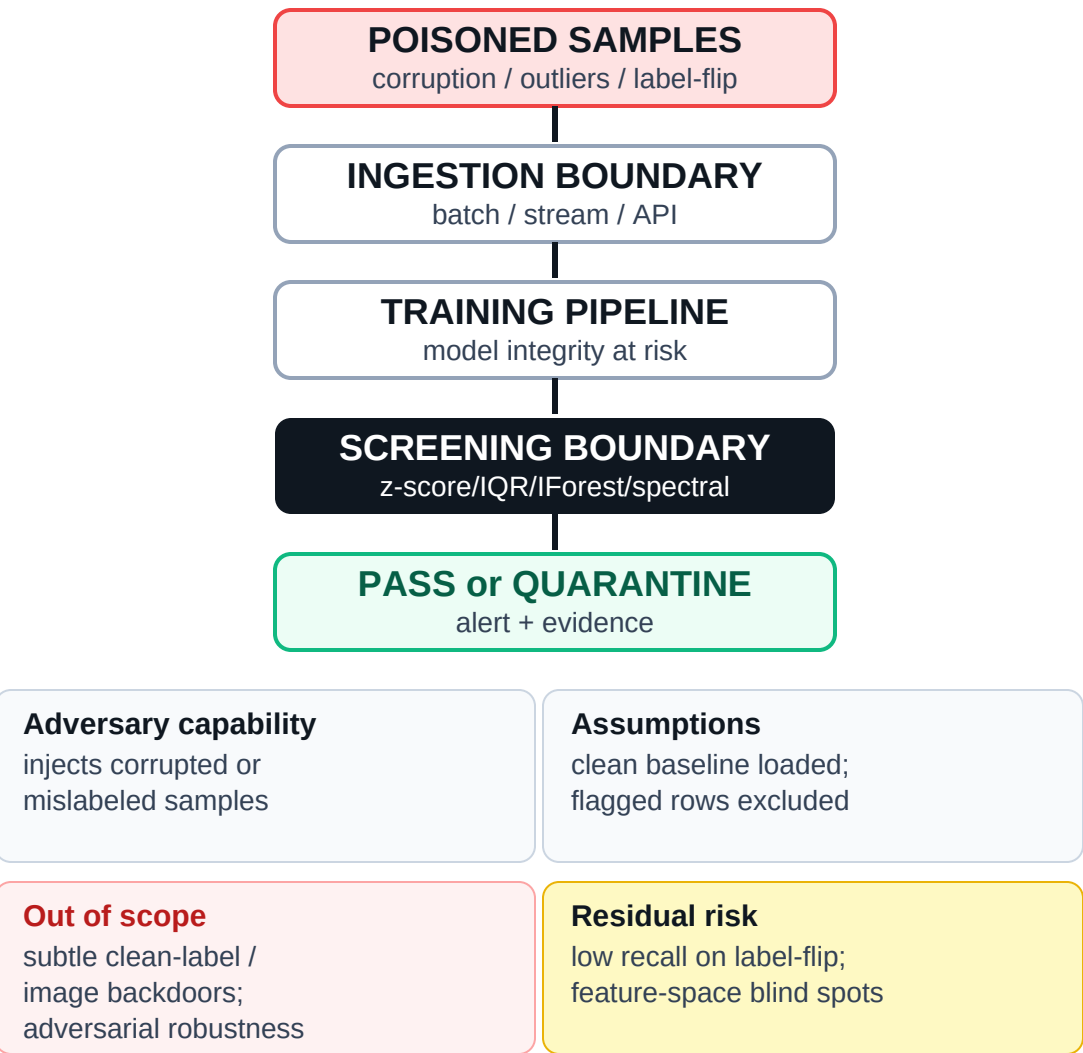
RESEARCH QUESTION

How effectively can statistical screening identify poisoned or corrupted training samples before they enter the training pipeline?

01 • THE FAILURE MODE

Poisoned or corrupted samples enter training silently. A vendor feed drifts or a shared table gets bad rows, and precision quietly drops over retraining cycles — the pipeline never errors, so nobody checks the data. A screening layer at ingestion (MITRE ATLAS AML.T0020) can catch gross corruption — but not subtle attacks.

02 • THREAT MODEL

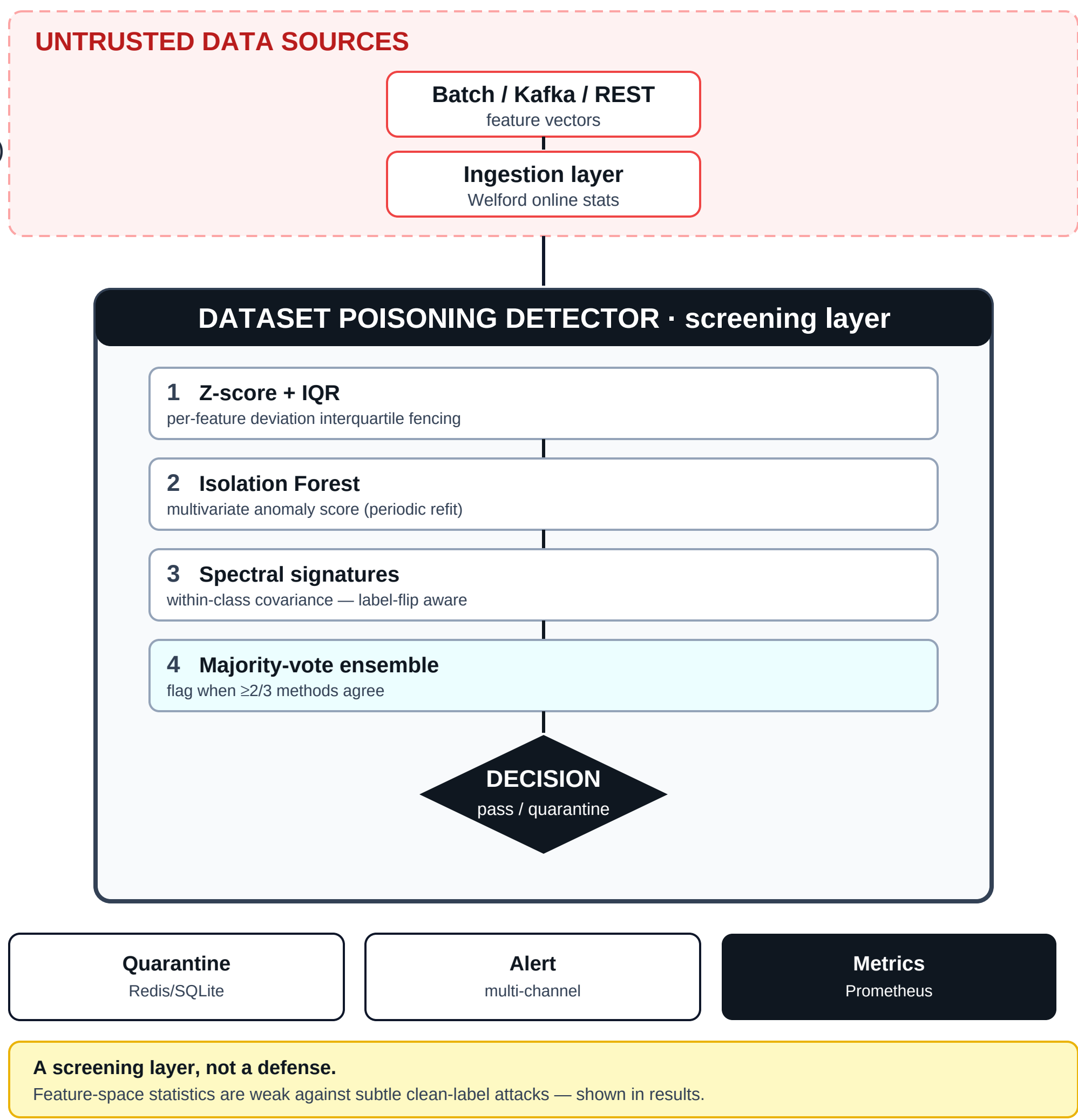


03 • OBJECTIVE

Determine how well an unsupervised statistical ensemble screens ingestion data — and measure explicitly where it fails.

- 01 – Z-score + IQR + IsolationForest
- 02 – Spectral signatures (label-aware)
- 03 – Majority-vote ensemble + quarantine
- 04 – Streaming (Welford) + drift

HERO • INGESTION SCREENING ENSEMBLE



RESULTS • WHERE SCREENING FAILS (measured)

0.37
BEST F1 (spectral)

0.08
WORST F1 @5%

Label-flip benchmark, seed 2018, 2000×100 synthetic. F1 rises with contamination but stays low — by design honest.

Spectral F1 vs contamination	
@ 5% poison	0.08
@ 10% poison	0.23
@ 20% poison	0.37

SPECTRAL vs ENSEMBLE F1 (@20%)

Spectral F1 (x100)	37%
Ensemble F1 (x100)	23%
Spectral wins	67%

Spectral beats ensemble on 2/3 rates; both remain weak on label-flip. spectral_benchmark.json

BENCHMARK PROVENANCE

results/spectral_benchmark.json (committed). Tran et al. 2018 setup. Throughput = environment-scoped microbenchmark; no production SLA claimed.

NOT ESTABLISHED BY THIS REPO

High recall on label-flip attacks. Detection of subtle clean-label / image backdoors. Robustness guarantee.

SECURITY BOUNDARY

PROTECTS / DETECTS

- gross outliers
- obvious corruption
- distribution shift
- replay (fingerprint)
- provenance logging

DOES NOT ESTABLISH

- subtle clean-label attacks
- image backdoors
- targeted poisoning
- robustness proof

Figure 1. Ensemble screening at the ingestion boundary; spectral analysis targets label-flip attacks feature-space

METHOD • EFFICACY BENCHMARK

1 Synth data
make_classification

2 Inject
label-flip

3 Score
4 methods

4 Vote
ensemble

5 Measure
P/R/F1

6-7 Compare
spectral vs ens

SCREENING ≠ DEFENSE

CONTRIBUTION

An ingestion-boundary statistical screening ensemble with a reproducible efficacy benchmark that quantifies its own weakness: feature-space methods cannot reliably detect label-only corruption, and spectral analysis only partially recovers it. Honesty is the contribution.

LIMITATIONS

- 1. Low recall on label-flip (F1 0.08–0.37 measured).
- 2. Feature-space methods blind to clean-label attacks.
- 3. Streaming stats are per-feature; correlations lost.
- 4. Synthetic benchmark ≠ real pipeline data.
- 5. First-pass filter, not adversarial robustness.

FUTURE WORK

- Evaluate on representative real pipeline data.
- Label-aware detectors beyond spectral.
- Multivariate streaming statistics.
- Clean-label attack detection research.
- Calibrated thresholds per dataset.

CONCLUSION

The tool screens ingestion data with a 4-method ensemble and quarantines outliers. Its committed benchmark deliberately shows weak label-flip F1 (0.08–0.37), documenting exactly where statistical screening breaks down. It is a filter, not a defense.

REPRODUCE
pytest tests/
python benchmark/cifar10_label_flip_benchmark.py